

Sherry Wilson

Senior Python & Django Backend Developer

Colchester, England, UK | +44 7403 408990 | sherrywilly521@gmail.com | linkedin.com/in/sherry-wilson-python-dev | Right to work in the UK (Graduate Route visa, valid until 2027)

PROFESSIONAL SUMMARY

Senior Python and Django Backend Developer with 4+ years of hands-on experience building scalable REST and GraphQL APIs, asynchronous processing systems, and cloud-ready backend platforms. Proven record of leading a 3-person backend team, delivering systems used by thousands of users, and driving measurable outcomes: 30% throughput improvement, 25% latency reduction, 30% faster media delivery, and 15% revenue uplift across enterprise client platforms. MSc Artificial Intelligence & Its Applications, University of Essex.

KEY ACHIEVEMENTS

- Led backend delivery for major client platforms including ITC (Classmate), Fabelle, Greencell Mobility (Neugo), and Nippon Paints.
- Improved backend throughput by **30%** through asynchronous processing with Celery, Redis, and Flower across production systems.
- Reduced system latency by **25%** via Kafka-based microservices architecture and service optimisation across production platforms.
- Supported a **15% revenue uplift** through scalable backend architecture and platform reliability improvements for high-profile clients.
- Built HLS video streaming pipelines that improved media delivery speed by **30%** for eLearning platforms at scale.
- Reduced frontend-server data load by **30%** through optimised GraphQL endpoint design.

CORE COMPETENCIES

Backend & APIs: Django, Django REST Framework, FastAPI, Flask, GraphQL, REST API design

Cloud & DevOps: AWS EC2, Docker, Nginx, load balancers, microservices deployment

AI & Automation: LangChain, RAG architecture, Gemini, LLM integration, Pandas, Matplotlib

Async & Messaging: Celery, Redis, Flower, Kafka, background processing

Data & Storage: PostgreSQL, SQL optimisation, ChromaDB

Delivery: Team leadership, mentoring, architecture design, API testing, stakeholder communication

PROFESSIONAL EXPERIENCE

Senior Python Backend Developer

Aug 2021 – Jul 2024

Deepsense Digital Solutions — Chennai, India

- Led a team of 3 backend developers delivering production systems for ITC (Classmate), Fabelle, Greencell Mobility (Neugo), and Nippon Paints.
- Designed and built scalable backend architectures with Django and Django REST Framework, serving thousands of users across multiple client platforms.
- Built and optimised GraphQL endpoints, reducing frontend-server data load by 30%.

- Implemented asynchronous task processing with Celery, Redis, and Flower, improving throughput by 30% and increasing system observability.
- Tuned PostgreSQL queries and database flows for stable, high-throughput performance under peak traffic conditions.
- Deployed AWS EC2, Docker, Nginx reverse proxy, and load-balanced infrastructure for high-availability production releases.
- Integrated Kafka-based microservices communication patterns, reducing system latency by 25%.
- Implemented adaptive HLS streaming with FFmpeg for eLearning platforms, improving media delivery speed by 30%.
- Contributed to a 15% revenue uplift through scalable backend architecture and platform reliability improvements.

AI Intern

Jul 2025 – Aug 2025

Victoria Solutions — London Area, UK

- Performed end-to-end data analysis and cleaning with Pandas for AI model training datasets, improving data quality and model-readiness.
- Built Matplotlib-based reporting dashboards to communicate model and dataset insights to stakeholders and research leads.
- Prepared and structured large datasets for ML pipelines, accelerating downstream feature engineering and model training workflows.

Python LLM Developer

Sep 2025 – Oct 2025

AquSag Technologies — Remote

- Analysed and triaged GitHub issues across open-source Python libraries to identify high-impact engineering challenges for LLM evaluation benchmarks.
- Assessed unit test coverage and repository code quality to benchmark LLM performance in real-world bug-fixing and code-generation scenarios.
- Collaborated with AI researchers to curate issue sets and repository benchmarks for rigorous LLM engineering capability assessment.
- Provided technical guidance to junior engineers on code quality, testing standards, and delivery practices.

Freelance Python Backend Developer

Jun 2020 – Jul 2021

Self-employed — Kerala, India

- Delivered custom Django, Flask, and FastAPI backend solutions for client projects requiring security, scalability, and clean API design.
- Built Celery and Redis background job workflows to automate client operations and reduce manual processing overhead.
- Containerised and deployed services with Docker across multiple environments.
- Integrated ML functionality into production systems through secure REST APIs.

SELECTED PROJECTS

Document Assistant API — RAG-powered FastAPI service using LangChain, ChromaDB, and Gemini to answer questions from PDF collections with citation-grounded responses.

HLS Video Streaming Platform — Adaptive HLS delivery pipelines with FFmpeg, AWS S3, and Nginx-backed serving; improved media delivery speed by 30% at scale.

Multi-Tenant E-Commerce Backend — Django DRF and GraphQL backend services with Celery, Redis, Kafka, PostgreSQL, and Docker for multi-client retail operations serving thousands of users.

ML Data Pipeline & Dashboard — Automated ingestion, cleaning, feature engineering, and reporting with Python, Pandas, and Matplotlib for AI model workflows.

EDUCATION

Master of Science — Artificial Intelligence & Its Applications Jan 2024 – Jan 2025
University of Essex, Colchester, UK

Bachelor of Computer Applications — Computer Programming & Specific Applications 2018 – 2021
University of Kerala, India

CERTIFICATIONS

- FastAPI — The Complete Course 2025 (Beginner + Advanced)
- Advanced Django
- Full UK Driving Licence

ADDITIONAL EXPERIENCE

Team Leader Jan 2026 – Present
Welshwood Manor — Colchester, UK

Senior Healthcare Assistant / Care Assistant Jul 2024 – Jan 2026
Care Circle — Colchester, UK

These roles reflect recent UK-based employment while continuing to develop backend and AI projects during the transition into the local technology market.